LESSON 3.4

Determine Equivalent Ratios Using Graphs

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can graph the values from a ratio table as points on the coordinate plane.

WHY THIS MATTERS The organizers of a soccer league ordered 6 of the bags shown to hold soccer balls. Each bag holds the same number of soccer balls. How many soccer balls will the bags hold?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Coordinate plane *Spanish: Plano cartesiano*	A grid with a line going across and a line going up to plot points.	x across, y up
Ordered pair *Spanish: Par ordenado*	Two numbers (x, y) that tell where a point is on a grid.	(2, 14)
Linear pattern *Spanish: Patrón lineal*	Points that make a straight line on a grid.	(1,2), (2,4), (3,6) all line up
Proportional *Spanish: Proporcional*	Two amounts that grow together at the same rate.	A straight line from (0,0) through (2,6) and (4,12)
Origin *Spanish: Origen*	The point (0, 0) where the two grid lines cross.	0 pages, 0 stickers
Ratio table *Spanish: Tabla de razones*	A table of equal ratios that shows a pattern.	Cups of juice: 2, 4, 6 | Cups of water: 3, 6, 9

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Each tray holds 8 muffins (x = trays, y = muffins). Write and plot three points.

1. Table: 1 tray = 8, 2 trays = 16, 3 trays = 24.
2. Pairs: (1, 8), (2, 16), (3, 24).
3. Plot each pair from (0, 0): across x, then up y.

Answer: The points lie on one straight line through (0, 0)

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

A recipe uses 1 cup of sugar for every 3 cups of flour.

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The point (_____ , _____) means _____ .
- The origin means _____ .
- I found y by multiplying x by _____ .
- The points form a straight line because _____ .

Word bank coordinate plane · ordered pair · origin · x-axis · y-axis · plot · point · straight line · table · multiply · across · up

WATCH OUT!

A common mistake

A common mistake in Graph Ratio Tables is writing the ordered pair in the wrong order

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Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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Answer: The points lie on one straight line through (0, 0)

1 **Circle the letter of the best answer. Show how you know.**

A ratio table shows (2, 6), (4, 12), and (6, 18). Which ordered pair comes next if the pattern continues?

(A) (7, 20)
(B) (8, 24)
(C) (8, 22)
(D) (10, 24)

STEPS TO TRY

1. Watch each coordinate separately: x goes 2, 4, 6, … and y goes 6, 12, 18, …
2. Find the jump in the x-values and the jump in the y-values, then apply both jumps to (6, 18).
3. The x-values climb by 2, so add 2 to the 6 first. Then find the y-jump from 6 → 12 → 18 and add it to 18.

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

When you graph equivalent ratios, the points will always form what shape?

(A) A curved line

(B) A straight line through the origin

(C) A zigzag

(D) A circle

STEPS TO TRY

1. Think about what happened when you plotted equivalent-ratio points earlier in this lesson.
2. Equivalent ratios grow steadily: every time x rises by the same amount, y rises by the same amount. What shape is steady growth?
3. Picture plotting (1, 2), (2, 4), (3, 6) and connecting them — and check whether (0, 0) would fit the pattern too.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

A ratio table shows (2, 8), (3, 12), and (5, 20). What is the y-value when x = 7?

(A) 21

(B) 24

(C) 28

(D) 35

STEPS TO TRY

1. Compare each pair: 2 → 8, 3 → 12, 5 → 20. What is done to x to get y every time?
2. Find the multiplier by dividing y by x for any pair — it should be the same for all three. Then apply it to x = 7.
3. Divide 8 ÷ 2 to find the rule's multiplier. Then multiply 7 by that number.

My work — one step per line	What I did to both sides

1. **Look** *What is done to the variable?*
2. **Undo** *Do the opposite — both sides.*
3. **Simplify** *One side at a time.*
4. **Check** *Put it back in.*

4 **Circle the letter of the best answer. Show how you know.**

Which point would NOT be on the graph of the ratio 1:5?

(A) (2, 10)

(B) (3, 10)

(C) (4, 20)

(D) (5, 25)

STEPS TO TRY

1. On the 1:5 graph, every y-value is exactly 5 times its x-value.
2. Test each point: multiply its x by 5 and see whether you land on its y. You're hunting for the one that misses.
3. Work through the points in order, multiplying each x by 5. The one whose y does not match is the answer.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

What pattern do the plotted points make? How does the graph show the ratio of 1 sundae to 2 ounces of sauce?

SENTENCE STARTER Every time the sundaes go up by ___, the ounces go up by ___, so the points line up in a ___ starting from ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help